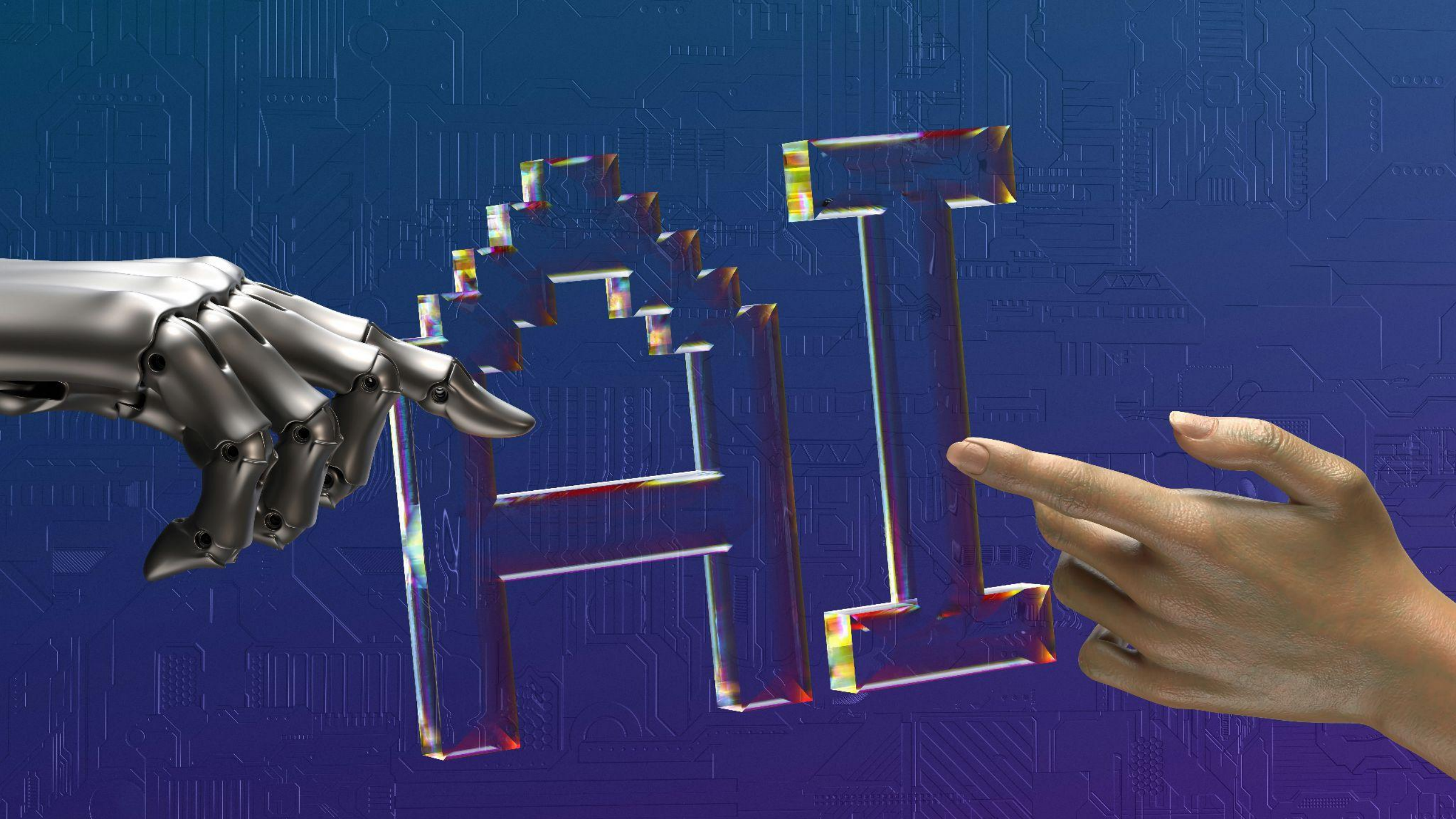


LLMs in technical writing

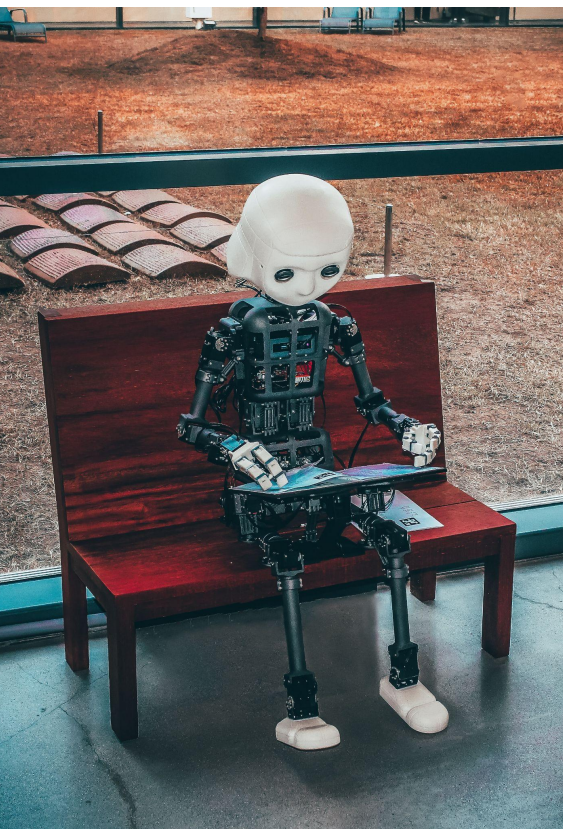
The fundamentals of technical writing | VUT 2026

The Red Hat Customer Content Services team



What we'll discuss today

- AI The Good & The Bad
- LLMs in Technical Writing
- Decoding Documentation
- AI Toolkit
- Prompt Engineering
- Refactoring Documentation
- Why Tech Writing Is Not Just Writing



How do you use AI tools, and which ones?



Why write docs when you have AI?

How AI empowers the technical writers

- Enhanced efficiency
- Automation
- Reduced human error
- Reduced physical risks
- Improved customer experience
- Democratization of skills



Why humans stay in the driver's seat

- Hallucinations
- No existing data
- Legal limitations
- Security and privacy concerns
- Environmental impact
- De-skilling

LLM use cases in Tech Writing



- Automated content auditing: “Identify sentences written in passive voice...”
- Find and update specific errors: “Update all UI instructions: 'Submit' is now 'Confirm'...”
- Description of code: “Explain the logic of this code to a junior...”
- Structural transformation: “Rewrite product documentation into an engaging blog post...”



Exercise I.

Decoding documentation

A sample GDB session



<https://www.sourceware.org/gdb/current/onlinedocs/gdb.html/Sample-Session.html>



Exercise I.

Analyze: What does this documentation demonstrate? What is GDB? Why would someone want to use it? What does this documentation consist of? Is it easy to understand it?

Extract the following:

- Main commands used
- The specific bug that was found
- How the bug was fixed

Present: Explain the bug and the fix in plain English

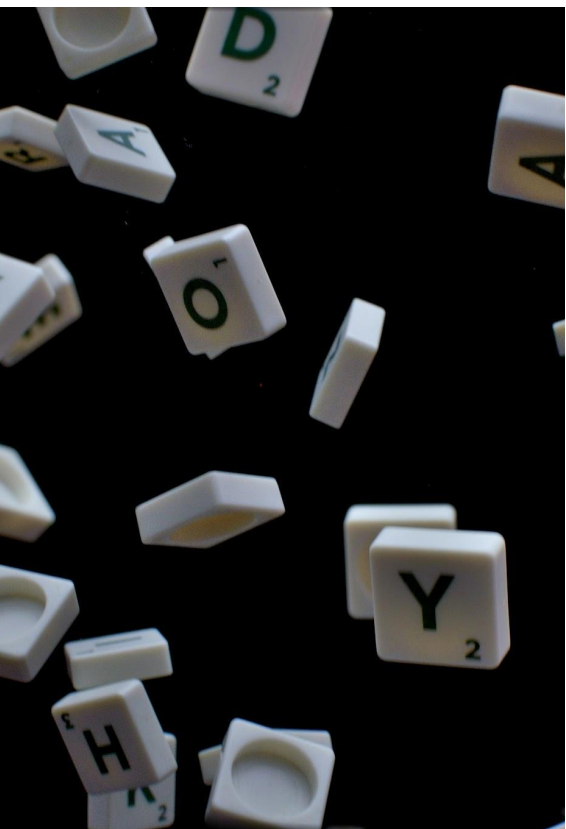


Tools that help us to write

- Chats: Copy-paste, isolated context
- Single-File CLI: Quick, targeted terminal edits
- Repo Agents: Multi-file, autonomous editing
- CI/CD Pipeline Jobs: Automated reviews of PRs



Prompt Engineering



How to write prompts for LLM?

"Write a README for my python project."



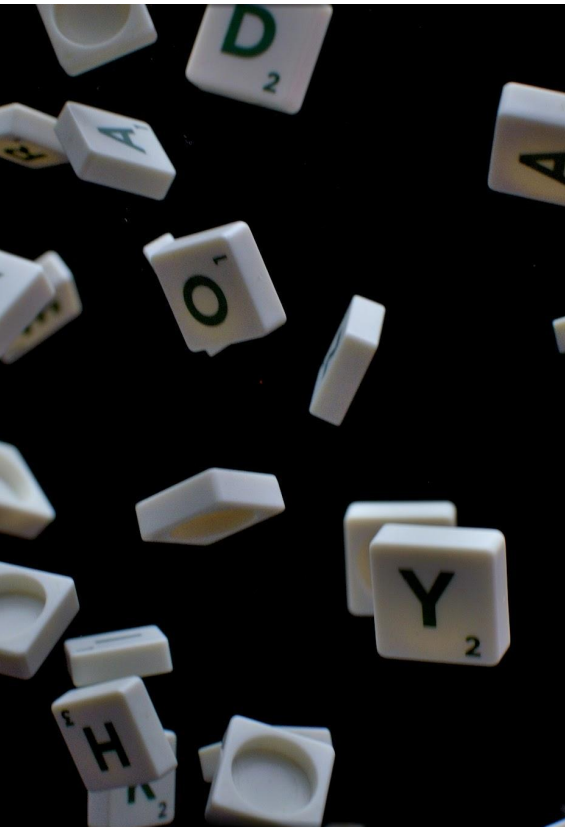
How to write prompts for LLM?

Act as an expert technical writer. Write a README.md for a Python CLI tool called 'LogSpy' that monitors server logs for errors.

Context: It's built for junior DevOps engineers. It requires Python 3.10+ and can be installed via `pip install logspy`.

Structure: Include an Introduction, Installation section, and a Quick Start code block showing how to run `logspy --path /var/log`.

Tone: Clear, direct, and authoritative. Avoid marketing fluff or buzzwords.



Exercise II.

Refactor documentation

A sample GDB session

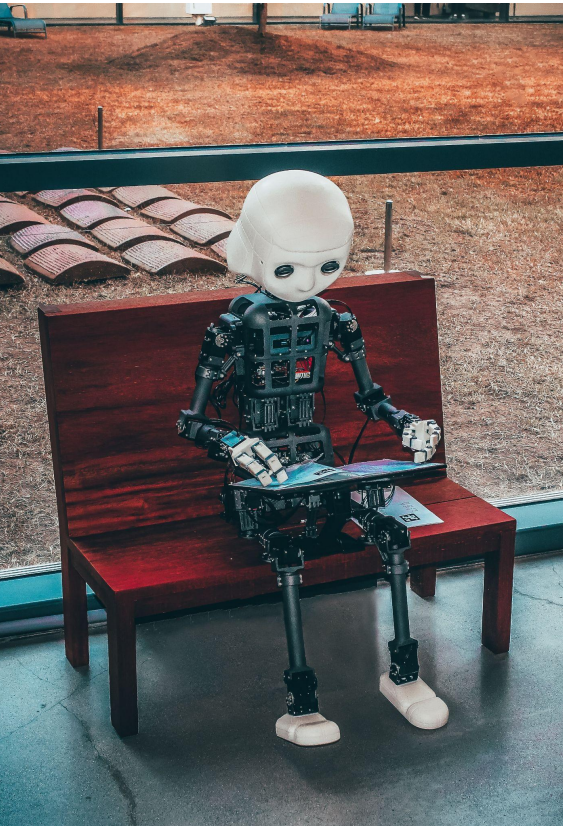


<https://www.sourceware.org/gdb/current/onlinedocs/gdb.html/Sample-Session.html>

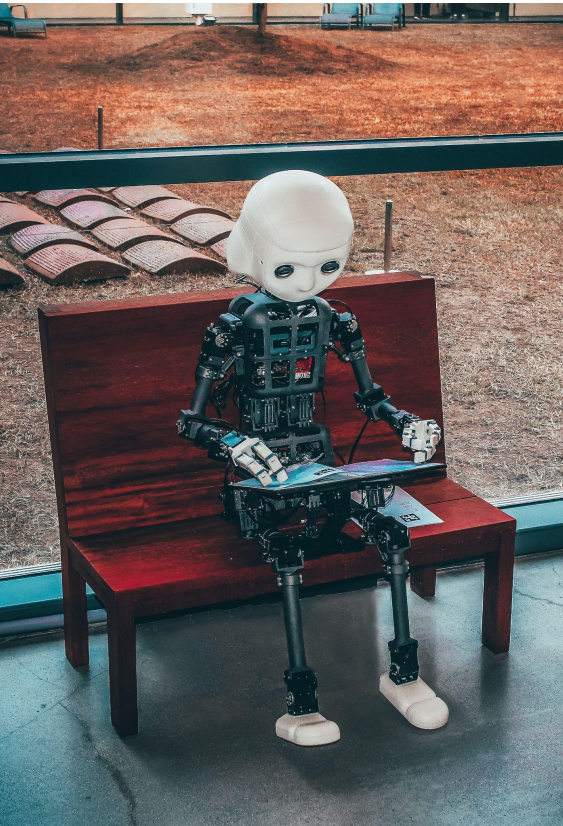


Technical writing is not just writing

- Research of new topics
- Planning, content strategy, user/content journey
- Verification and testing
- User experience



How can AI help students



Exploring your needs as a learner

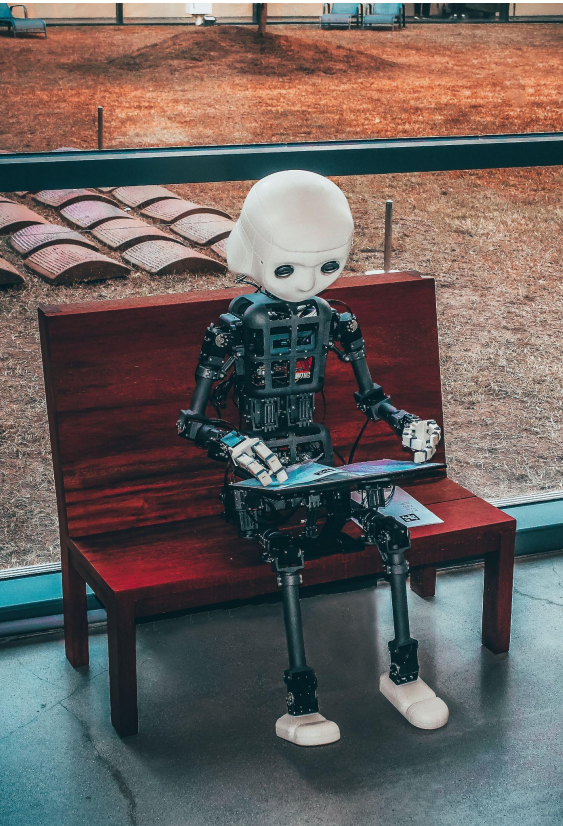
1. Clarify your position as a learner:

Academic context:

- What are you studying and at what level (year, specific courses)?
- What are your strongest and weakest subject areas?
- What types of assignments do you typically work on?
- What are your academic goals this term and beyond?

Learning style and challenges:

- How do you learn best (visual, verbal, hands-on, etc.)?
- What aspects of learning do you find most challenging?
- Where do you typically need the most help or support?
- What motivates you to learn and persist through difficulties?



Context

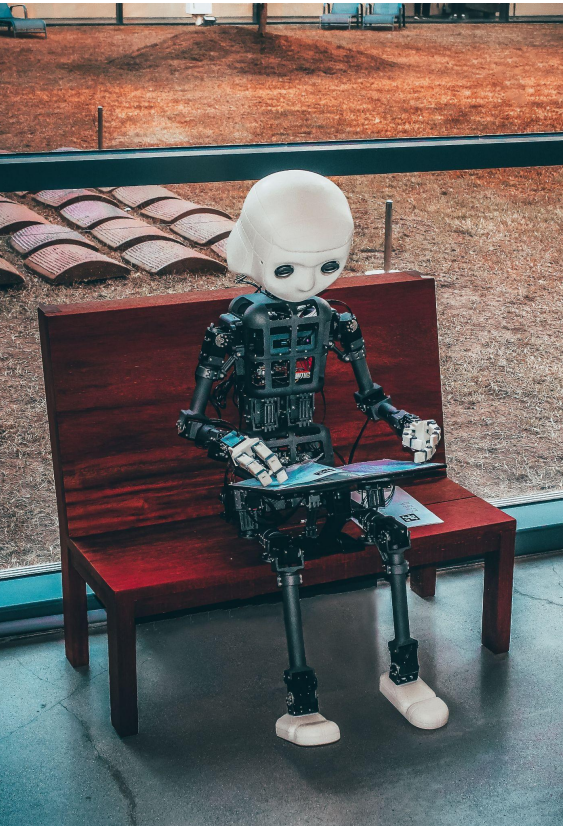
2. Create your learning context document with AI

Opening the conversation:

- Explain that you're a student building a learning context document
- Tell the AI this will help establish how you can work together effectively on academic tasks
- Make it clear that you want to use AI to enhance your learning, not replace it

Key areas to explore with the AI:

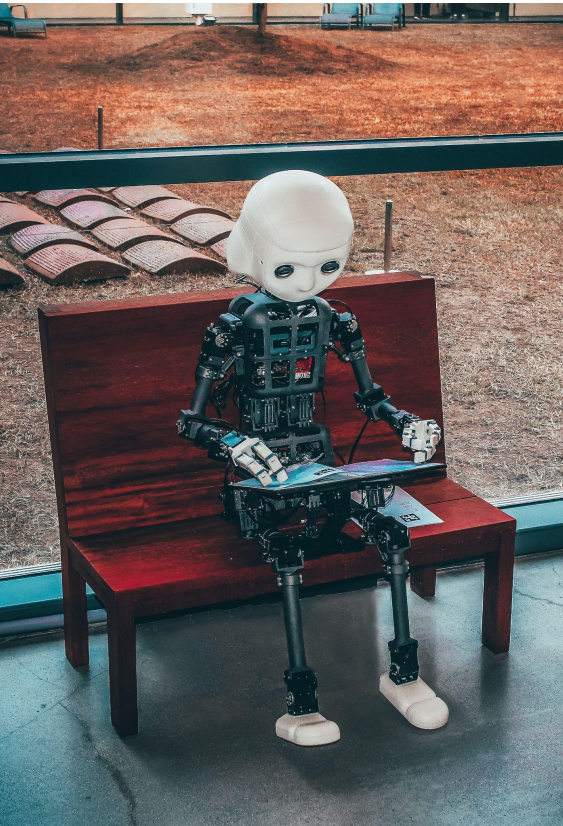
- Your current courses and learning challenges
- Your goals and what success looks like for you
- Your preferred study methods and what helps you understand concepts
- Situations where you want AI support versus where you want to work independently
- Your school's AI policies and how to work within them



Boundaries

Important boundaries to establish:

- Make it clear you want to learn, not have work done for you
- Discuss how you'll maintain academic integrity
- Establish when you'll work with AI versus independently
- Set expectations for the type of help you want (guidance, practice, feedback, etc.)



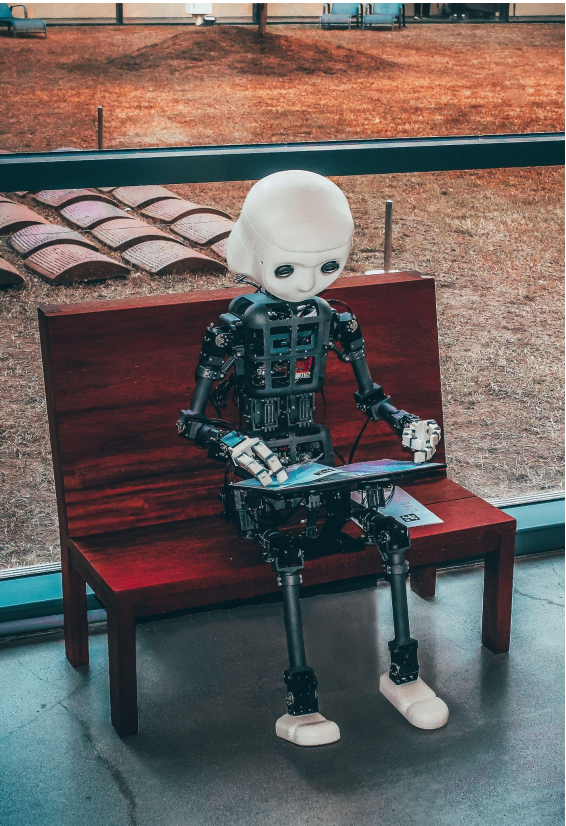
Reference document

3. Creating your reference document:

- Ask the AI to synthesize your discussion into a structured learning context document
- Review it together and make any necessary adjustments
- Include a section on "How I Want to Work with AI" based on your boundaries
- Save this document to share at the start of future AI learning sessions

Building the document together:

- Share the reflections from Step 1 with the AI
- Let the AI ask follow-up questions to understand your context better
- Be specific about examples of assignments or concepts you're working on
- Discuss what kind of AI support would actually help you learn versus just complete tasks



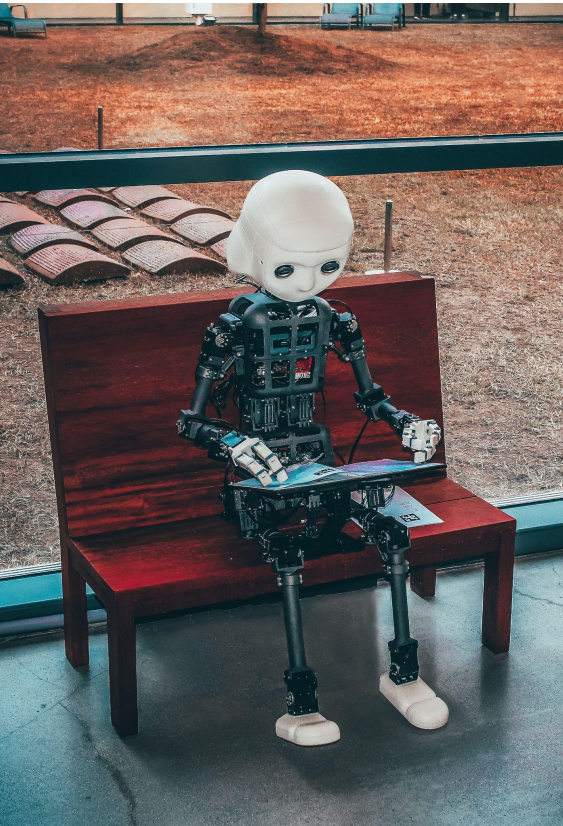
AI as a learning partner

Setting up your learning partner:

- If starting fresh, share your learning context document from the earlier slide
- Explain that you want to configure the AI as a study buddy for ongoing use
- Specify that you want help learning, not completing assignments

Establishing the learning partner's role: Work with the AI to define how it should help you learn:

- Ask it to always start by understanding what you're trying to learn before helping
- Request that it asks you questions rather than giving direct answers
- Have it check your understanding before moving to new concepts
- Ask it to provide practice problems that build on what you're studying
- Request that it points out connections to things you've learned before



Specific approaches for different needs

- For problem-solving: "Guide me with hints and questions, don't solve it for me"
- For concept review: "Test my understanding with progressively harder questions"
- For exam prep: "Quiz me and explain why wrong answers are wrong (and why correct answers are correct)"
- For writing: "Help me develop my own arguments through questioning"
- For reading comprehension: "Ask me to explain key concepts in my own words"
- For general planning: "Gather information from me about my various commitments so I can effectively plan out coursework and other activities I need to complete/attend"
- For specific assignment planning: "Help me test my understanding of this assignment brief so I am sure I understand what is expected of me"

Thank you



- Access documentation
- Download a PDF file



Exercise

gemini.google.com



Takeaways

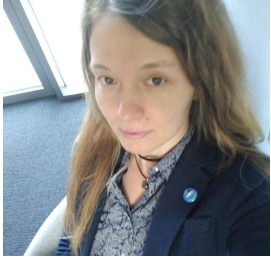
- Docs should inform AI, not the other way around
- Automation can be used cheaper and faster than AI in some cases
- **AI cannot think**, only predict the next token. You need to have the following skills: user focus, minimalism, stylistics, applying a style guide



Learn prompt engineering

[Prompt engineering guide](#)

About the authors:



Alexandra Nikandrova

Technical Writer for Red Hat

Former Devops



Jana Vrbkova

Technical Writer for Red Hat

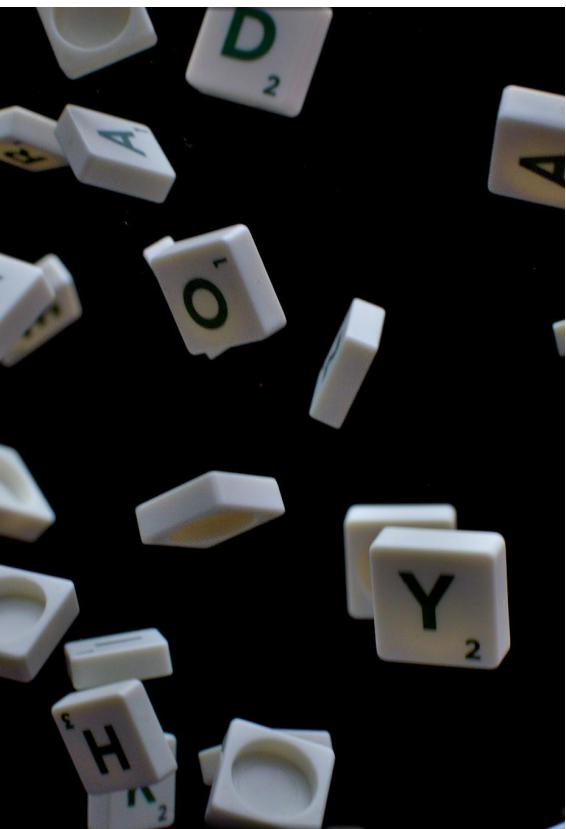
Favorite AIs: Google AI on Pixel

DeepL



Srikanth R

Technical Writer for Red Hat



How do large language models work?

- LLMs have trillions of parameters.
- How LLMs work:
 - **Data**
 - **Architecture**
 - **Training**
- They can be **fine-tuned** for specific tasks.
- They iteratively **predict the next token** based on context.



When to use AI

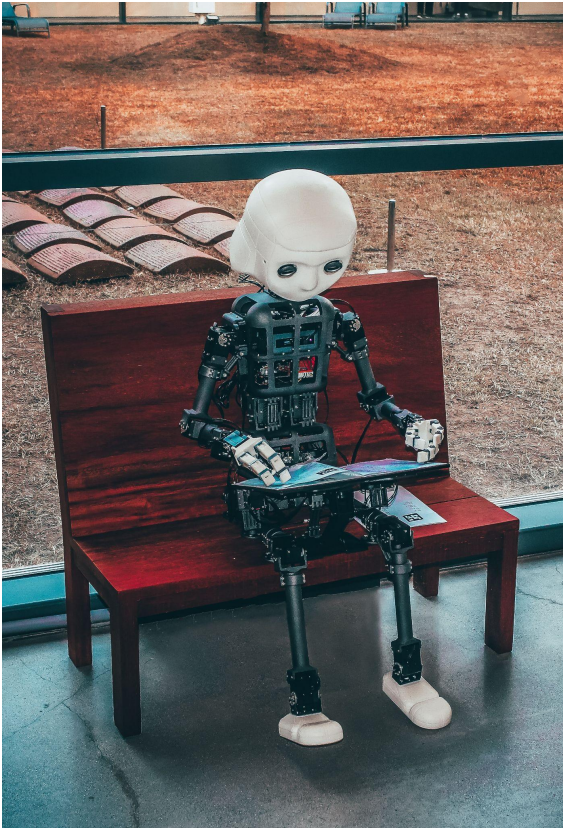
- Pattern recognition with no clear rules
- Processing natural language
- Modeling



When to use automation

- Predictable, repetitive rules (processing files, system monitoring)
- Low complexity
- Speed
- Limited data (no data required)
- Low cost (generally cheaper to implement)

Types of LLMs



- Architecture (encoder, decoder, mixture of experts)
- Training methodology (general purpose LLM, fine-tuned)
- Modality (text-only, multimodal)
- Licencing (proprietary, open-source)

Which text is written by a human TW?

A	B	C
Red Hat Enterprise Linux 9: GIMP 2.99.8 Technology Preview Available This release introduces GNU Image Manipulation Program (GIMP) version 2.99.8 as a Technology Preview in Red Hat Enterprise Linux 9. GIMP 2.99.8 delivers a comprehensive set of image editing and manipulation capabilities, enabling users to: • Perform basic image editing tasks: Crop, resize, adjust color and brightness, apply filters, and more. • Create and manipulate artwork: Draw vector graphics, paint with various brushes, and work with layers for complex compositions. • Automate repetitive tasks: Utilize scripting capabilities to automate workflows and save time. Important Note: GIMP 2.99.8 is provided as a Technology Preview. This means it is still under development and may contain bugs or stability issues. It is not intended for production use. This Technology Preview allows users to explore the capabilities of GIMP 2.99.8 before its official introduction in Red Hat Enterprise Linux 9. Feedback from users will be crucial in shaping the final offering. Installation: GIMP 2.99.8 is available through the Technology Preview repository. You can install it using the following command: <pre>sudo dnf install gimp-techpreview</pre> Disclaimer: Red Hat does not provide support for Technology Preview features. Use them at your own risk and discretion.	Feature: GNU Image Manipulation Program (GIMP) 2.99.8 Description: The GIMP package version 2.99.8 is now available as a Technology Preview in Red Hat Enterprise Linux 9. This pre-release version provides a glimpse into the upcoming GIMP 3 release. Users can explore and test this version using the RPM package for installation. Key Points: • Technology Preview Status: ◦ GIMP 2.99.8 is presented as a Technology Preview, showcasing features and enhancements that will be fully integrated into the official GIMP 3 release in future versions of RHEL 9. • Pre-Release Version: ◦ This package serves as a pre-release version, allowing users to experience and provide feedback on the latest developments in GIMP. Recommended Action: • Users interested in previewing the upcoming features in GIMP 3 can install the GIMP 2.99.8 package. • As a Technology Preview, this version is not intended for production use but offers users an opportunity to familiarize themselves with the upcoming GIMP release. Note: GIMP 2.99.8 is a forward-looking Technology Preview, giving users early access to the evolving capabilities of GIMP 3. Feedback from the community during this preview phase is valuable in refining and enhancing the user experience before the official release.	GIMP available as a Technology Preview in RHEL 9 GNU Image Manipulation Program (GIMP) 2.99.8 is now available in RHEL 9 as a Technology Preview. The <code>gimp</code> package version 2.99.8 is a pre-release version with a set of improvements, but a limited set of features and no guarantee for stability. As soon as the official GIMP 3 is released, it will be introduced into RHEL 9 as an update of this pre-release version. In RHEL 9, you can install <code>gimp</code> easily as an RPM package.

Exercise: Tips for prompting

Bonus: Custom instructions (system prompt)

Go to **Your profile > Customize ChatGPT** to customize the responses.

For example, create your own style guide:

1. **Be Clear:** Use simple and direct language. Avoid unnecessary jargon unless it's needed.
2. **Stay Formal:** Keep a formal and neutral tone. Don't add personal opinions or emotions.
3. **Talk to the Reader:** Address the reader directly with "you".
4. **Use Present Tense:** Describe actions or states in the present tense, not future tense.
5. **Use AsciiDoc:** Format text using AsciiDoc markup, not Markdown.
6. **Give Actions:** Provide clear instructions or guidance on what the reader should do.

Workshop: Automate or AI-ate?

- [TBA a couple specific scenarios from working on docs where AI could, but perhaps should not, be used]

Types of LLMs

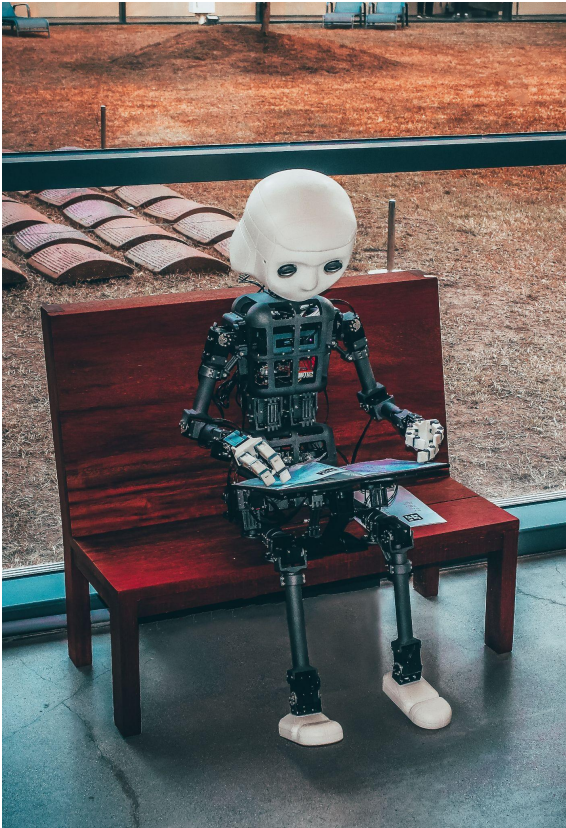
Open-source vs Proprietary

Open-source LLMs are publicly available, allowing for free use, modification, and distribution, enabling extensive customization.

Examples: Llama 3, Falcon, WizardLM, and so on.

Proprietary LLMs have limited access and customization options, often offered with strong performance and robust support.

Examples: OpenAI's GPT-4, Google's Gemini, Anthropic's Claude, and so on.



Types of LLMs

Encoder, decoder, mixture of experts

Encoder-only focus on understanding input data

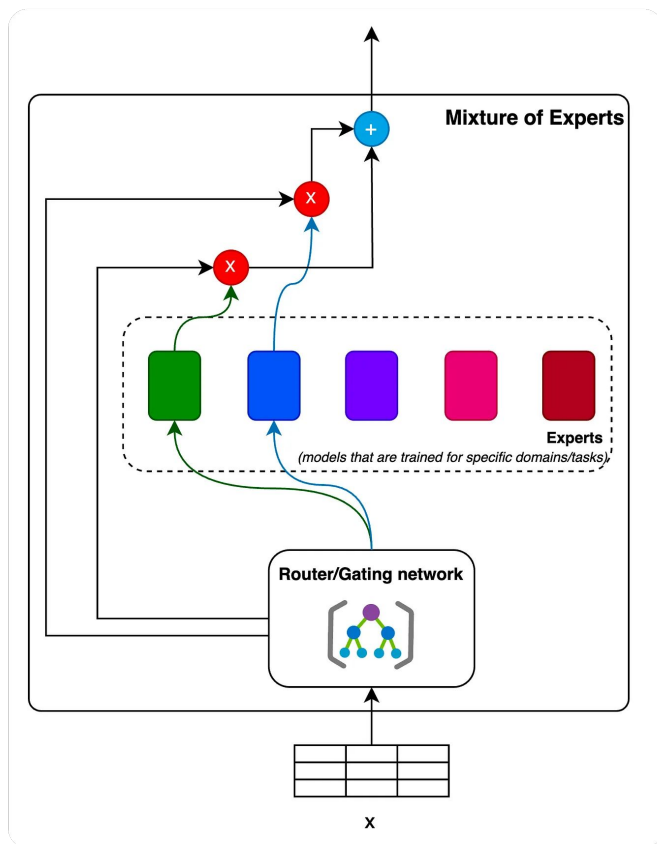
Examples: BERT (Bidirectional Encoder Representations from Transformers)

Decoder-only focus on generating context-aware text

Examples: GPT, LLaMA, Gemma, Claude

Mixture of Experts (MoE) focus on scalability and efficiency of all the models involved

Examples: Qwen3 (30B A3B)



Types of LLMs

General purpose vs Fine-tuned



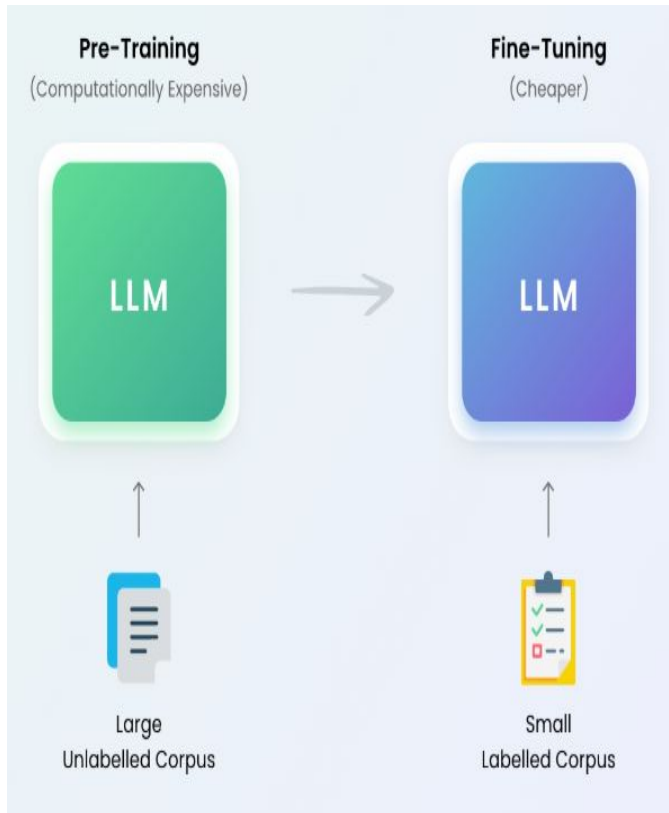
General purpose LLMs are trained on a wide range of data—can handle most tasks but not optimized for one domain.

Examples: ChatGPT, Claude, Gemini

Fine tuned LLMs are Specialized on specific domains like legal, medical, or technical documentation.

Examples: Writer.com (custom LLM), CodeLlama (for code), BioGPT (for biomedical content)

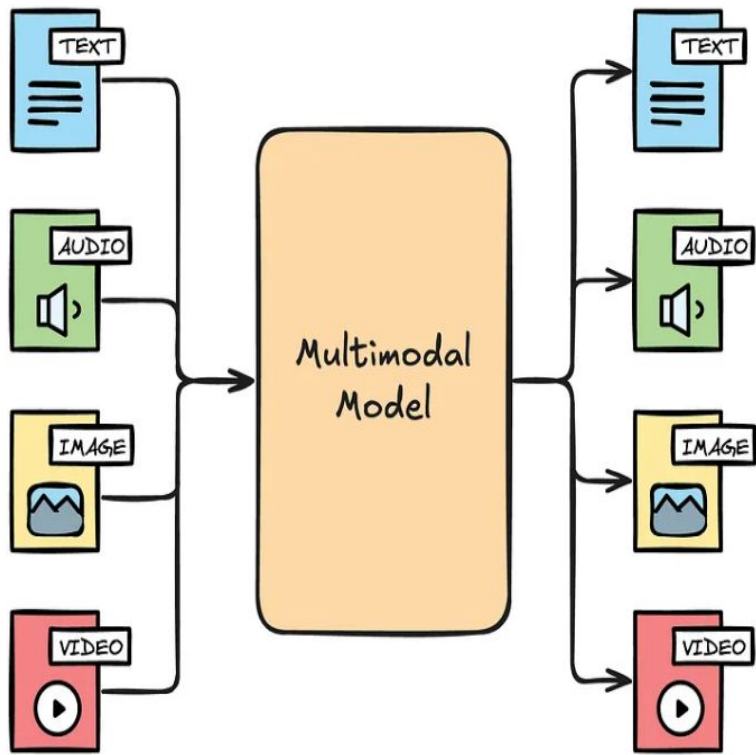
Types of LLMs



Retrieval-Augmented Generation (RAG)

General purpose model fine-tuned to your needs.

Types of LLMs



Text-only vs Multi-modal

Text-only LLMs are primarily designed to understand, analyze, and generate text-based content.

Examples: GPT-3, LLaMA, BLOOM.

Use Cases: Text generation, translation, summarization, question answering, chatbots.

Multi-modal LLMs are capable of processing and generating content across multiple data types, such as text, images, audio, and video.

Examples: Flamingo (DeepMind), Kosmos-1 (Microsoft), LLaVA (open-source), PaLM-E (Google).

Use Cases: Computer Vision, Audio-Visual AI, Robotics, Content Creation, and so on.

EXERCISE: Now you try it!

Cause: Too restricted checks in openssl

Consequence: random tls connection failures

Workaround (if any): just reconnect

Result: handshake succeeds

gemini.google.com

bing.com/chat

chat.openai.com

Exercise: Tips for prompting

"Document how to manage and configure passkey authentication devices in the Identity Management (IdM) environment."

- Provide contextual details.
- Be clear and specific in your prompts.
- Focus on one aspect at the time (content format, structure, language).
- Work one section at a time (heading, abstract, procedural steps, verification...).
- Provide examples of the style or tone you're aiming for.
- Refine the prompts using different wording to get better results.
- Feed ChatGPT with content or reference material.